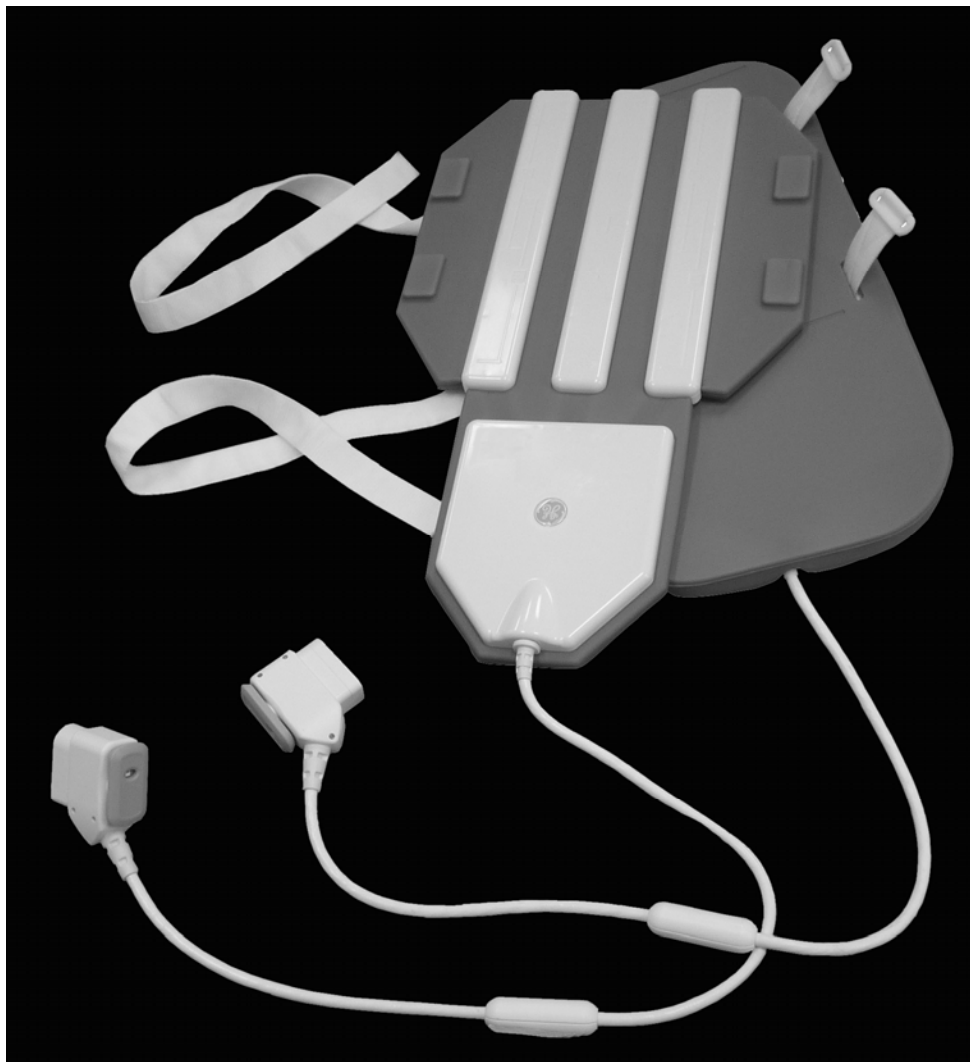


GE Healthcare

3.0T 32-Channel Torso Array Coil

Service Manual



NC004SVM

Rev 4

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警告 (ZH-CN)	本维修手册仅提供英文版本。 - 如果客户的维修服务人员需要非英文版本，则客户需自行提供翻译服务。 - 未详细阅读和完全理解本维修手册之前，不得进行维修。 - 忽略本警告可能对维修服务人员、操作人员或患者造成电击、机械伤害或其他形式的伤害。
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UPOZORENJE (HR)	Ovaj servisni priručnik dostupan je na engleskom jeziku. - Ako davatelj usluge klijenta treba neki drugi jezik, klijent je dužan osigurati prijevod. - Ne pokušavajte servisirati opremu ako niste u potpunosti pročitali i razumjeli ovaj servisni priručnik. - Zanemarite li ovo upozorenje, može doći do ozljede davatelja usluge, operatera ili pacijenta uslijed strujnog udara, mehaničkih ili drugih rizika.
VÝSTRAHA (CS)	Tento provozní návod existuje pouze v anglickém jazyce. - V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka. - Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah. - V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.
ADVARSEL (DA)	Denne servicemanual findes kun på engelsk. - Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse. - Forsøg ikke at servicere udstyret uden at læse og forstå denne servicemanual. - Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk stød, mekanisk eller anden fare for teknikeren, operatøren eller patienten.

WAARSCHUWING (NL)	Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar. - Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan. - Probeer de apparatuur niet te onderhouden alvorens deze onderhoudshandleiding werd geraadpleegd en begrepen is. - Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.
WARNING (EN)	This service manual is available in English only. - If a customer's service provider requires a language other than english, it is the customer's responsibility to provide translation services. - Do not attempt to service the equipment unless this service manual has been consulted and is understood. - Failure to heed this warning may result in injury to the service provider, operator or patient from electric shock, mechanical or other hazards.
HOIATUS (ET)	See teenindusjuhend on saadaval ainult inglise keeles - Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest. - Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist. - Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.
VAROITUS (FI)	Tämä huolto-ohje on saatavilla vain englanniksi. - Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla. - Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen. - Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.
ATTENTION (FR)	Ce manuel d'installation et de maintenance est disponible uniquement en anglais. - Si le technicien d'un client a besoin de ce manuel dans une langue autre que l'anglais, il incombe au client de le faire traduire. - Ne pas tenter d'intervenir sur les équipements tant que ce manuel d'installation et de maintenance n'a pas été consulté et compris. - Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.
WARNUNG (DE)	Diese Serviceanleitung existiert nur in englischer Sprache. - Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es Aufgabe des Kunden für eine entsprechende Übersetzung zu sorgen. - Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben. - Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendiensttechnikers, des Bedieners oder des Patienten durch Stromschläge, mechanische oder sonstige Gefahren kommen.
ΠΡΟΕΙΔΟΠΟΙΗΣΗ (EL)	Το παρόν εγχειρίδιο σέρβις διατίθεται μόνο στα αγγλικά. - Εάν ο τεχνικός σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει τις υπηρεσίες μετάφρασης. - Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό αν δεν έχετε συμβουλευτεί και κατανοήσει το παρόν εγχειρίδιο σέρβις. - Αν δεν προσέξετε την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στον τεχνικό σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.

FIGYELMEZTETÉS (HU)	Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el. - Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészítése. - Ne próbálja elkezdni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték. - Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.
ÁDVÖRUN (IS)	Þessi þjónustuhandbók er aðeins áanleg á ensku. - Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálþjónustu. - Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin. - Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.
AVVERTENZA (IT)	Il presente manuale di manutenzione è disponibile soltanto in lingua inglese. - Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione. - Procedere alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto. - Il mancato rispetto della presente avvertenza potrebbe causare lesioni all'addetto alla manutenzione, all'operatore o ai pazienti provocate da scosse elettriche, urti meccanici o altri rischi.
警告 (JA)	このサービスマニュアルには英語版しかありません。 - サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。 - このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。 - この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。
경고 (KO)	본 서비스 매뉴얼은 영어로만 이용하실 수 있습니다. - 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우, 번역 서비스를 제공하는 것은 고객의 책임입니다. - 본 서비스 매뉴얼을 참조하여 숙지하지 않은 이상 해당 장비를 수리하려고 시도하지 마십시오. - 본 경고 사항에 유의하지 않으면 전기 쇼크, 기계적 위험, 또는 기타 위험으로 인해 서비스 제공자, 사용자 또는 환자에게 부상을 입힐 수 있습니다.
BRĪDINĀJUMS (LV)	Šī apkopes rokasgrāmata ir pieejama tikai angļu valodā. - Ja klienta apkopes sniedzējam nepieciešama informācija citā valodā, klienta pienākums ir nodrošināt tulkojumu. - Neveiciet aprīkojuma apkopi bez apkopes rokasgrāmatas izlasīšanas un saprašanas. - Šī brīdinājuma neievērošanas rezultātā var rasties elektriskās strāvas trieciena, mehānisku vai citu faktoru izraisītu traumų risks apkopes sniedzējam, operatoram vai pacientam.
İSPÉJIMAS (LT)	Šis eksploatavimo vadovas yra tik anglų kalba. - Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, suteikti vertimo paslaugas privalo klientas. - Nemėginkite atlikti įrangos techninės priežiūros, jei neperskaitėte ar nesupratote šio eksploatavimo vadovo. - Jei nepaisysite šio įspėjimo, galimi paslaugų tiekėjo, operatoriaus ar paciento sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų.

ADVARSEL (NO)	Denne servicehåndboken finnes bare på engelsk. • Hvis kundens serviceleverandør har bruk for et annet språk, er det kundens ansvar å sørge for oversettelse. • Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått. • Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.
OSTRZEŻENIE (PL)	Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim. • Jeśli serwisant klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta. • Nie próbować serwisować urządzenia bez zapoznania się z niniejszym podręcznikiem serwisowym i zrozumienia go. • Niezastosowanie się do tego ostrzeżenia może doprowadzić do obrażeń serwisanta, operatora lub pacjenta w wyniku porażenia prądem elektrycznym, zagrożenia mechanicznego bądź innego.
AVISO (PT-BR)	Este manual de assistência técnica encontra-se disponível unicamente em inglês. • Se outro serviço de assistência técnica solicitar a tradução deste manual, caberá ao cliente fornecer os serviços de tradução. • Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. • A não observância deste aviso pode ocasionar ferimentos no técnico, operador ou paciente decorrentes de choques elétricos, mecânicos ou outros.
ATENÇÃO (PT-PT)	Este manual de assistência técnica só se encontra disponível em inglês. • Se qualquer outro serviço de assistência técnica solicitar este manual noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução. • Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. • O não cumprimento deste aviso pode colocar em perigo a segurança do técnico, do operador ou do paciente devido a choques eléctricos, mecânicos ou outros.
ATENȚIE (RO)	Acest manual de service este disponibil doar în limba engleză. • Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere. • Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service. • Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.
ОСТОРОЖНО! (RU)	Данное руководство по техническому обслуживанию представлено только на английском языке. • Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод. • Перед техническим обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения. • Несоблюдение требований данного предупреждения может привести к тому, что специалист по техобслуживанию, оператор или пациент получит удар электрическим током, механическую травму или другое повреждение.
UPOZORENJE (SR)	Ovo servisno uputstvo je dostupno samo na engleskom jeziku. • Ako klijentov serviser zahteva neki drugi jezik, klijent je dužan da obezbedi prevodilačke usluge. • Ne pokušavajte da opravite uređaj ako niste pročitali i razumeli ovo servisno uputstvo. • Zanemarivanje ovog upozorenja može dovesti do povređivanja serviser, rukovaoca ili pacijenta usled strujnog udara ili mehaničkih i drugih opasnosti.
UPOZORNENIE (SK)	Tento návod na obsluhu je k dispozícii len v angličtine. • Ak zákazníkovi poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka. • Nepokúšajte sa o obsluhu zariadenia, kým si neprečítate návod na obsluhu a neporozumiete mu. • Zanedbanie tohto upozornenia môže spôsobiť zranenie poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, mechanické alebo iné ohrozenie.

ATENCION

(ES)

Este manual de servicio sólo existe en inglés.

- Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual.
- No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio.
- La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.

VARNING

(SV)

Den här servicehandboken finns bara tillgänglig på engelska.

- Om en kunds servicetekniker har behov av ett annat språk än engelska, ansvarar kunden för att tillhandahålla översättningstjänster.
- Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken.
- Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.

DİKKAT

(TR)

Bu servis kılavuzunun sadece ingilizcesi mevcuttur.

- Eğer müşteri teknisyeni bu kılavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer.
- Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz.
- Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

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ATTENTION!

Installation of this coil
MUST be performed by a GE
field engineer. Please contact
your field engineer to arrange
for installation.

Attempts to perform installation of
this coil without proper system
configuration will result in system
errors.

1 3.0T 32-Channel Torso Array Coil Introduction

1.1 Product Identification

The 3.0T 32-Channel Torso Array Coil consists of an Anterior and a Posterior half. Each half is uniquely identified by using the labels below:

Illustration 1.1: Torso Coil Labels

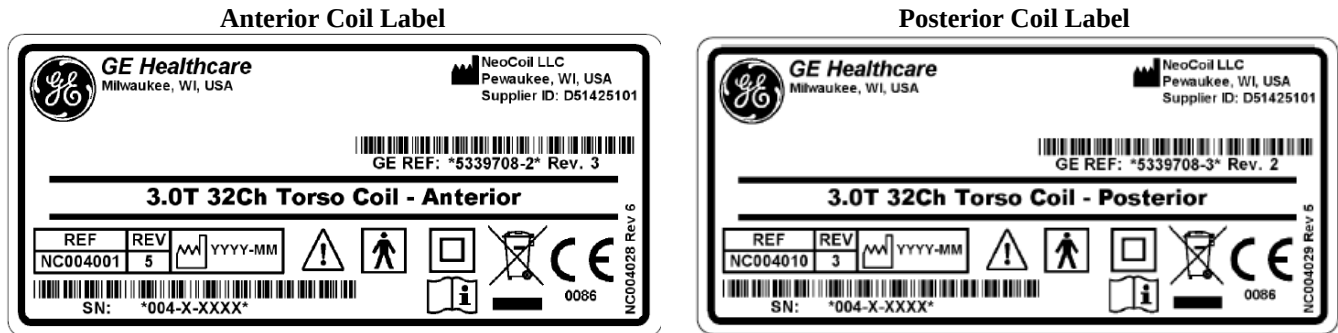


Illustration 1.2 shows images of each half of the 3.0T 32-Channel Torso Array Coil without pads or straps.

Illustration 1.2: The 3.0T 32-Channel Torso Array Coil without pads or straps.



1.2 Compatibility

The 3.0T 32-Channel Torso Array coil is compatible with GE 3.0T MRI scanners with P-Port connectors. The system must be configured for at least 32 channels.

1.3 Shipping Contents

1.3.1 *Shipping Contents 3.0T 32-Channel Torso Array coil*

Table 1.1: Box 1 Contents

Part Name	GE Part Number	NeoCoil P/N	Qty.
3.0T Torso Anterior Coil with Cable	5339708-2	NC004001	1
3.0T Torso Posterior Coil with Cable	5339708-3	NC004010	1
Strap Assembly	N/A	NC004034	2
Instructions For Use (with CD)	N/A	NC004IFU	1
Coil Definition CD	5339708-7	N/A	1
Service Manual	N/A	NC004SVM	1
Quick Reference Guide	N/A	NC004089	1

Table 1.2: Box 2 Contents

Part Name	GE Part Number	NeoCoil P/N	Qty.
Phantom Positioner	5339708-5	NC004009	1
Chin Pad	N/A	NC004006	1
Anterior Patient Comfort Pad	N/A	NC004007	1
Cable Pad	N/A	NC004008	1
Posterior Patient Pad	N/A	NC004033	1

1.4 Related Documentation

3.0T 32-Channel Torso Array Instructions for Use, NeoCoil Part Number NC004IFU.

1.5 Environmental Requirements

Storage Requirements

Coil should be stored in the scanner room. The environmental conditions in which this coil may be stored when not in use are as follows:

	Min	Max
Temperature	-15°C	55°C
Humidity	20%	95%

Dimensions

Anterior: 36.2" L x 21.1" W x 1.5" H (91.9cm x 53.6cm x 3.8cm)

Posterior: 32.5" L x 17.3" W x 2.5" H (82.5cm x 43.9cm x 6.4cm)

Weight

Anterior: 10.3lbs. (4.7 kg)

Posterior: 15.5lbs. (7.0 kg)

2 3.0T 32-Channel Torso Array Coil Installation Procedure

2.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Timing Not Applicable	25 mins	30 min

2.2 Preliminary Requirements

2.2.1 *Tools and Test Equipment*

Item	Qty	Part#
Posterior Phantom Positioner	1	5339708-5
Legacy Phantom*	2	5341595 Alternate Part Numbers: 2381682-10 or 2406200 or 5325953-6
Unified Phantom*	2	5343347-2
Coil Software CD: Discovery MR750 Torso Coil Definitions Disc	1	5339708-7
DVD SaveInfo Media	1	5165713 (this is qty of 50 DVDs)

* Note: Required to use Legacy Phantom or Unified Phantom.

2.2.2 *Safety – Not applicable*

2.2.3 *Required Conditions*

Condition
The minimum software build to install this coil is 20.0_M4

2.3 Procedure

2.3.1 Make sure that the latest software build is loaded on the system prior to installing the Torso coil. The minimum software build is 20.0_M4; the 32Ch Torso protocols will only install if the software build is 20.0_M4.

Software 20.1_IB1 and later does not require installation. Proceed to step 2.3.7 to confirm the coil is installed on the system and perform the finalization steps. If the 32Ch Torso coil is not present in coil DB continue with the following steps for installation.

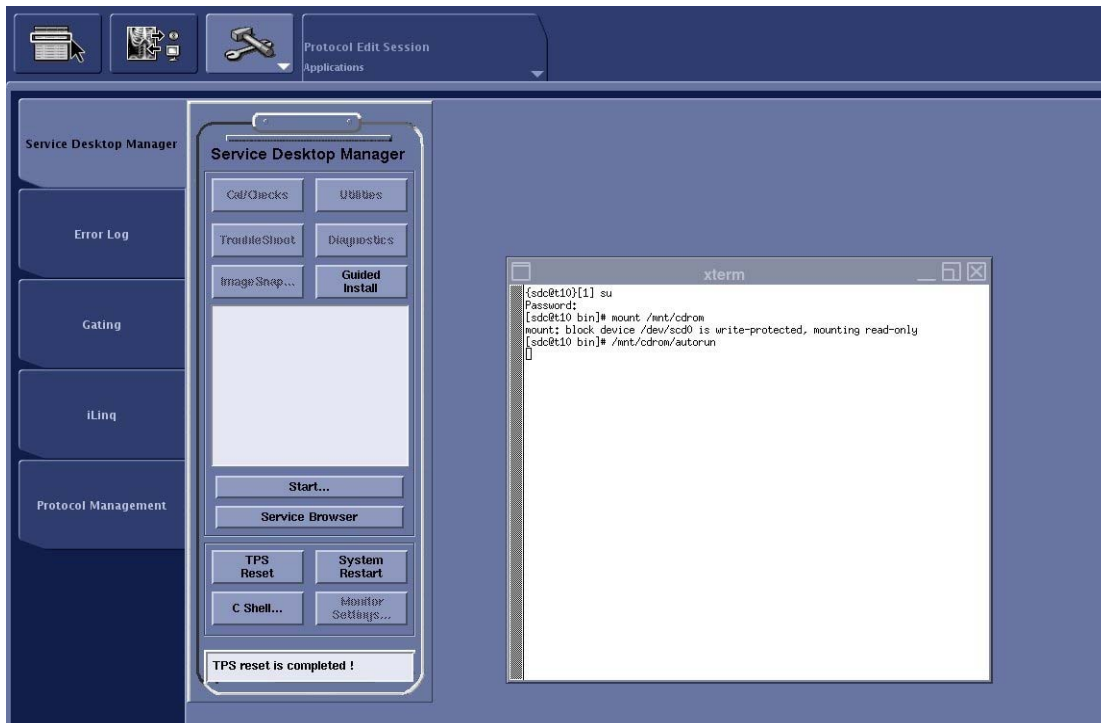
NOTE: Do not plug in the coil and use AutoCoil to install the coil. This method will not install the protocols.

2.3.2 Insert the Torso Coil Definitions Disc into the default (typically upper) host DVD drive.

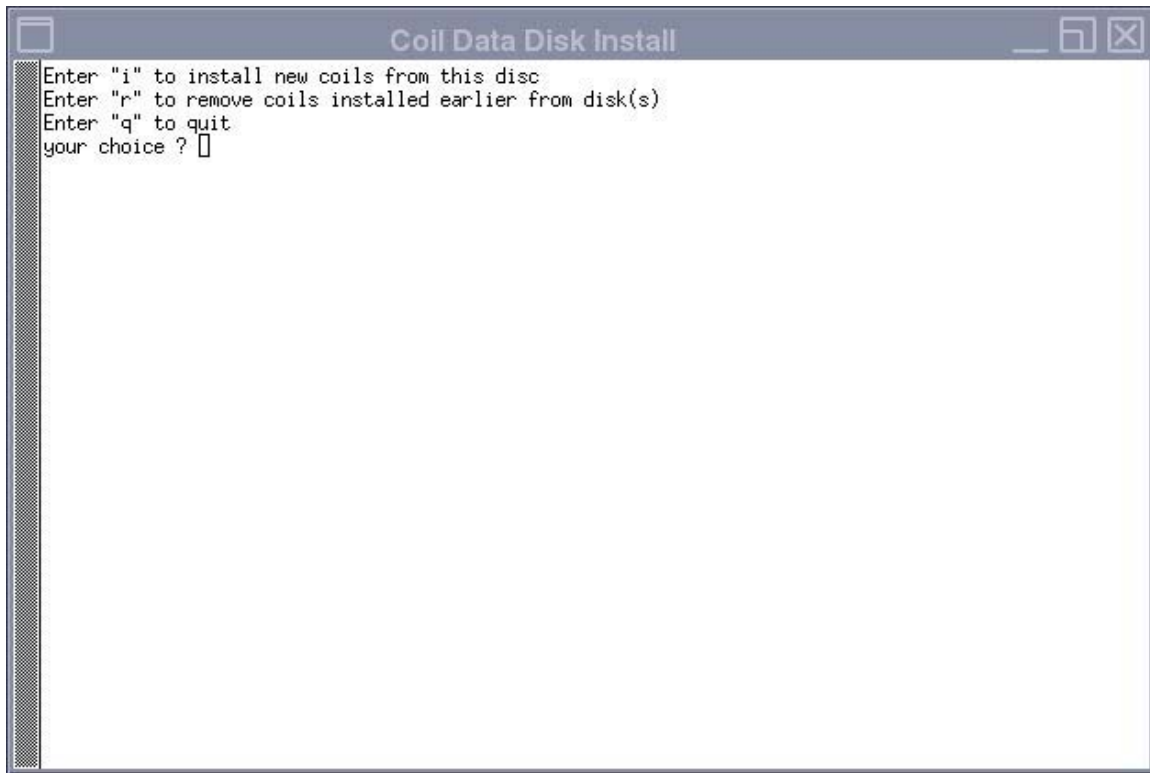
- 2.3.3 Open a C-Shell from the Service Desktop manager and type the following commands as shown in Illustration 2.1:

```
su
<root password>
mount /mnt/cdrom
/mnt/cdrom/autorun
```

Illustration 2.1: Installing Coil CD from Service Desktop

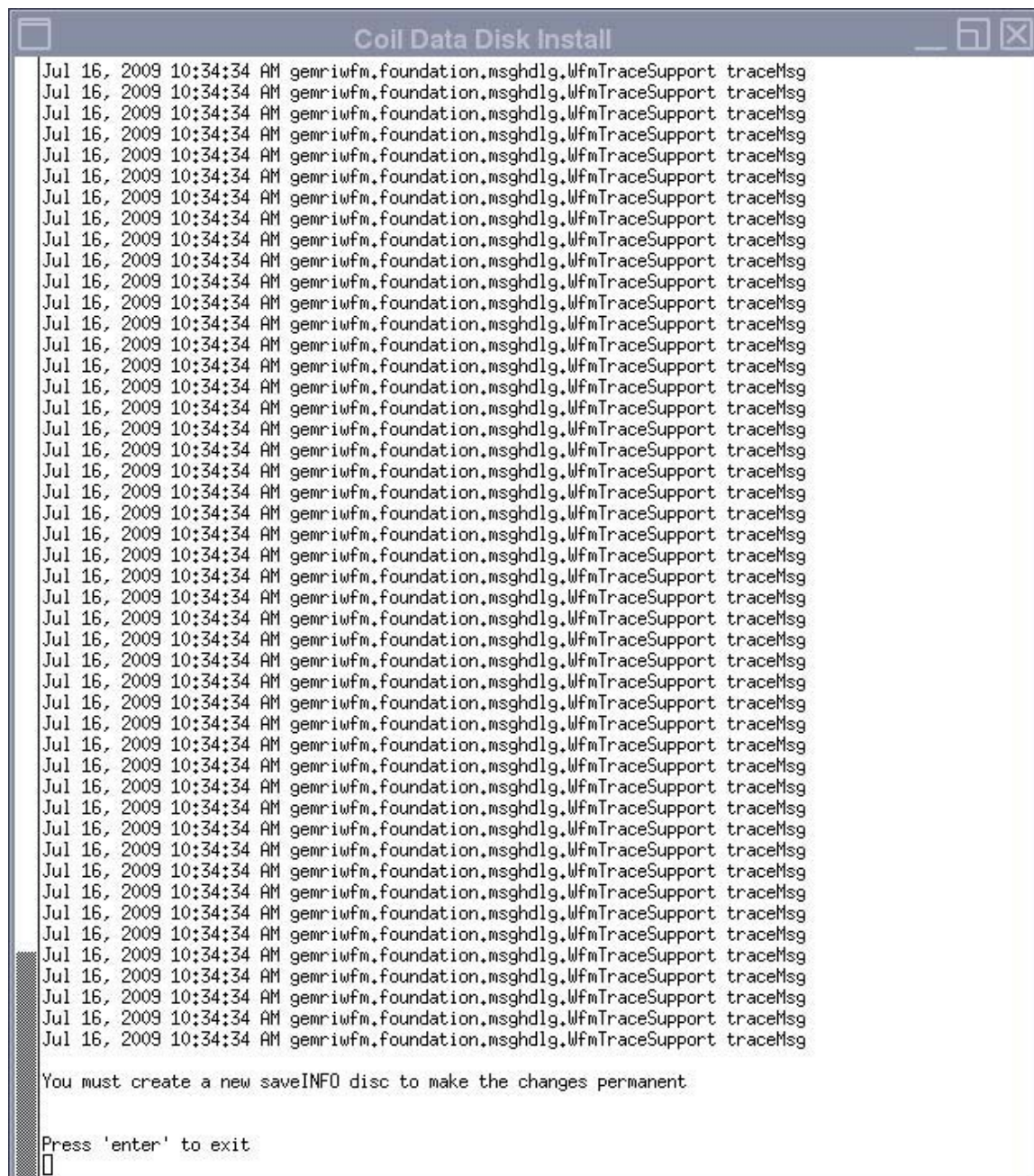


- 2.3.4 A Coil Data Disk Install window will open as shown in Illustration 2.2. Type “i” to install the CD contents.

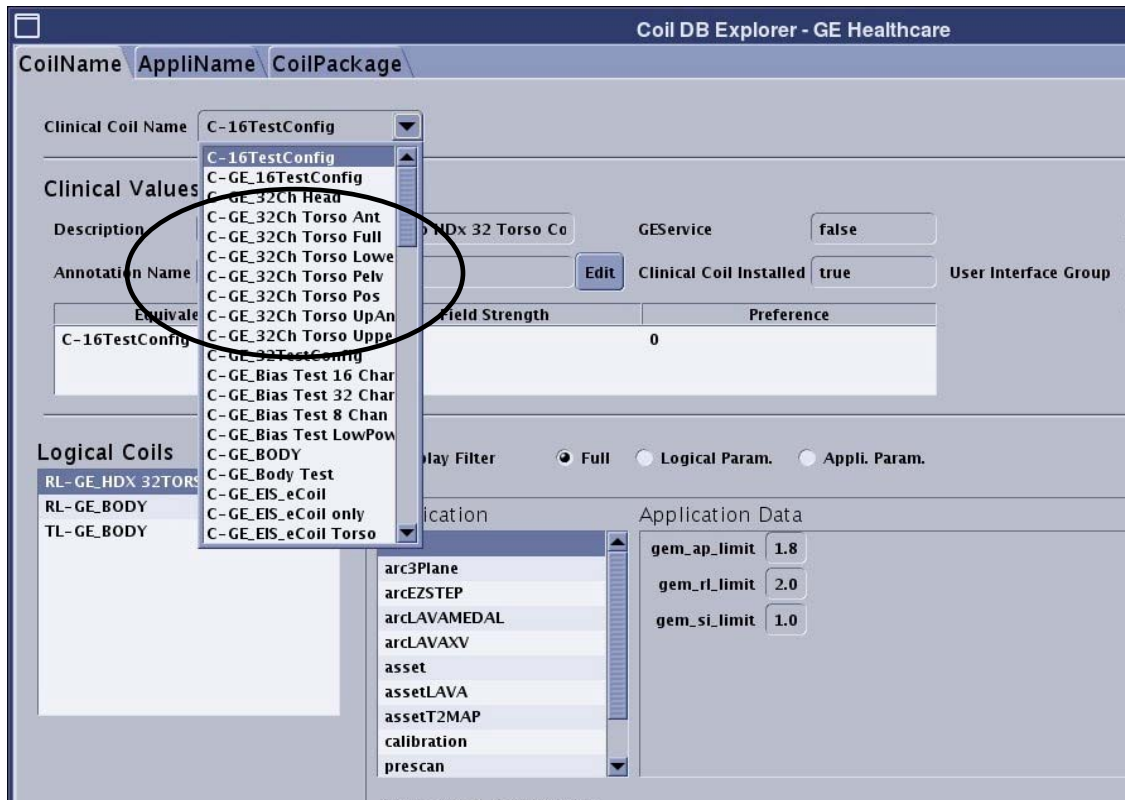
Illustration 2.2: CD Data Disk Install Window

- 2.3.5 When the contents of the CD installation are complete the cursor will appear and you will see the message as shown in Illustration 2.3. Press enter to exit the Coil CD installation.

Illustration 2.3: CD Contents Installation Complete



- 2.3.6 Type <eject> in the original C-Shell. Remove the CD from the host DVD drive.
- 2.3.7 To confirm the coil has been properly installed open the Service Browser from within the Service Desktop Manager. In the Service Browser, click on the Configuration tab, then the Coil Database Explorer under the configuration on the left.
- 2.3.8 To open the Coil Database Explorer select “Click here to start this tool”. The Coil DB Explorer window will open and under the “Coil Name” tab there is a “Clinical Coil Name” control. Click on this and there will be a dropdown list of all the logical coil names. There should be seven names that start with “C-GE_32ChTorso”. This indicates the coil is present in the Coil DB. Refer to Illustration 2.4.

Illustration 2.4: Confirmation of proper installation.

The presence of coil entries like those shown above is a confirmation that the coil installed properly. If you do not see the 32 Ch Torso coil modes, **DO NOT ATTEMPT TO INSTALL THE COIL MANUALLY, call the online center for additional information.**

2.3.9 Perform a SaveInfo under Guided Install to make the changes permanent.

2.4 Finalization

Perform the Multi-Coil Quality Assurance (MCQA) test with this coil. Refer to Section 3.

Reminder: Complete Product Locator information for serialized components. Failure to do this may result in failure of your site to receive future FMI's.

- 2.4.1 Go to GIB: http://gib.gehealthcare.com/gib/gib_MainMenu.jsp
- 2.4.2 After the log-in screen, click on [Global Installed Base entries]
- 2.4.3 Under "Serial Changes to Installed Base", enter the system ID of the system on which this coil is being installed. Click [Submit]
- 2.4.4 Click on [New Entries]
- 2.4.5 For both the Anterior and Posterior portions of the coil, enter the Model (Part number), Serial number, Installation Date and "A", then click [Submit].

3 3.0T 32-Channel Torso Array Coil MCQA Test

Follow this procedure to perform the MCQA test.

3.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Timing Not Applicable	10 mins/coil	Timing Not Applicable

3.2 Preliminary Requirements

3.2.1 Tools and Test Equipment

Item	Qty	Part#
Legacy Phantom*	2	5341595 Alternate Part Numbers: 2381682-10 or 2406200 or 5325953-6
Unified Phantom*	2	5343347-2
Phantom Positioner	1	5339708-5

* Note: Required to use Legacy Phantom or Unified Phantom.

3.3 Required Conditions

The following coil configuration files must be installed to run this tool.

- 32Torso Ant
- 32Torso Pos

3.4 Coil and Phantom Setup

Record the serial number of the coil(s) being used. The coil system cable must be directed towards the magnet and be plugged into port P1 or P2 for this test. The MCQA tool will be run separately for the posterior and anterior coils.

Table 3.1: Posterior Coil Setup

Visual	Description
	1. Remove any other surface coils (if present) from the cradle.

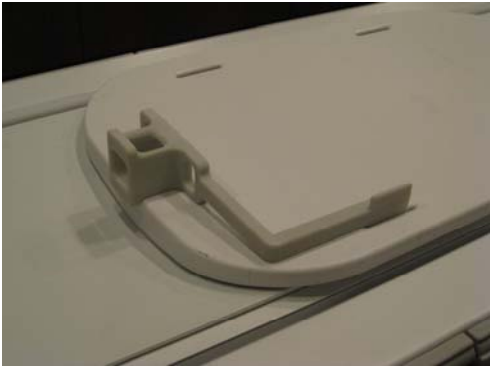
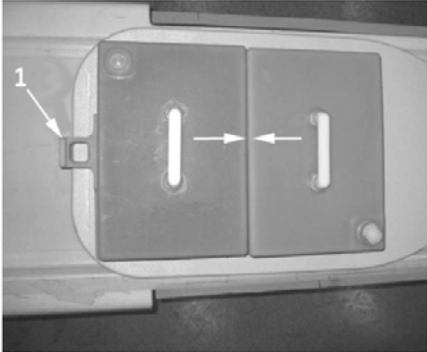
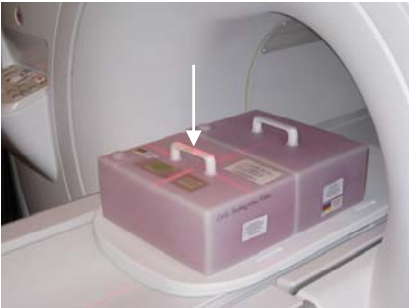
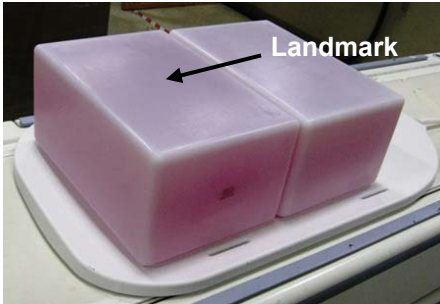
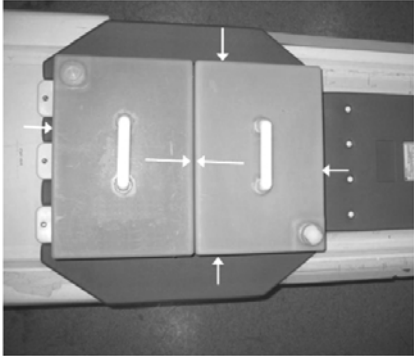
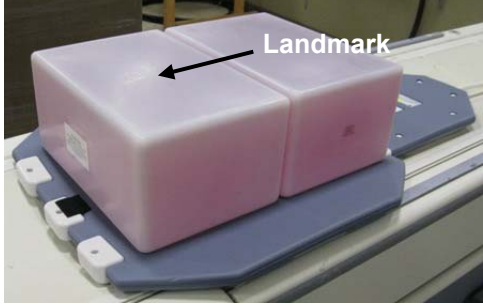
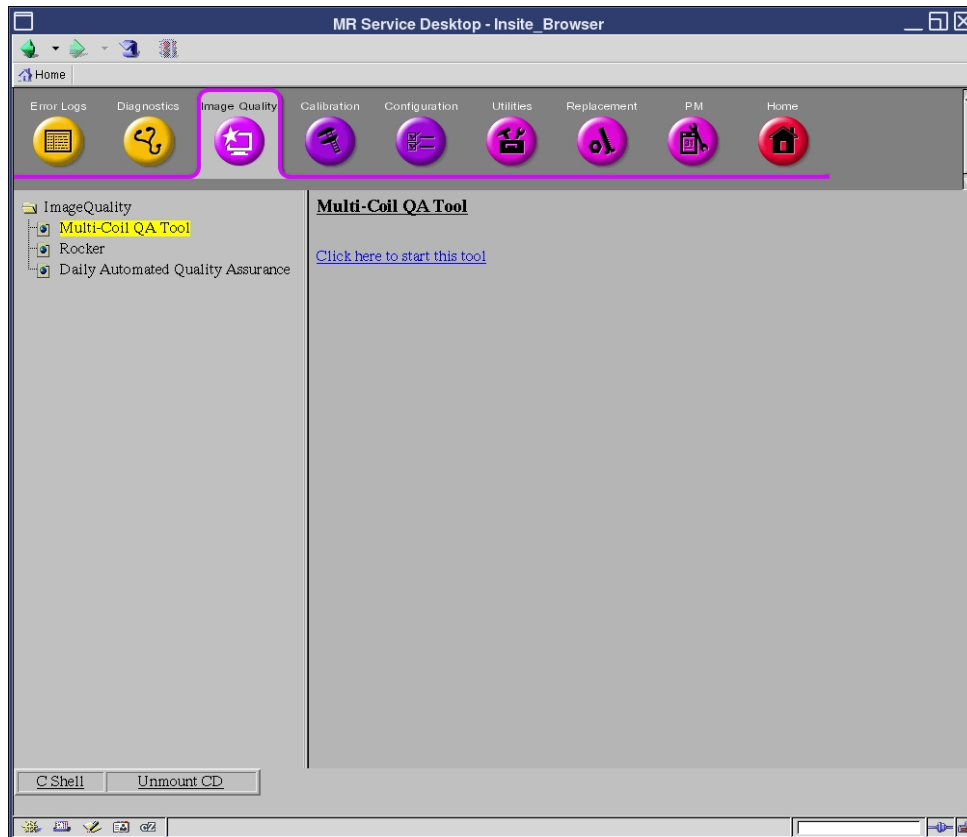
	2. Place the posterior coil and phantom positioner on the table as shown. Remove any pads present on the coil. The cable must be directed towards the magnet and be plugged into port P1 or P2.
 Legacy Phantom Setup  Unified Phantom Setup (one phantom shown)	3. Using the phantom positioner (1), center the phantoms in the R/L and S/I directions. Ensure that the phantoms are aligned with, and are touching each other.
 Legacy Phantom Landmark  Unified Phantom Landmark	4. Remove the phantom positioner. In the Legacy Phantom setup, landmark on the center of the phantom handle furthest from the coil port as shown. When using the Unified Phantoms, landmark the coil at the centre of the phantom that is away from the LPCA.

Table 3.2: Anterior Coil Setup

Visual	Description
 Legacy Phantom Setup  Unified Phantom Setup	1. Remove any other surface coils (if present) from the cradle. 2. Place the anterior coil and phantoms on the table as shown. The cable must be directed towards the magnet and be plugged into port P1 or P2. Ensure that the phantoms are centered in the R/L direction and also centered between the white plastic caps. The phantom positioner is not needed.
 Legacy Phantom Landmark  Unified Phantom Landmark	3. In the Legacy Phantom setup landmark on the center of the phantom handle furthest from the coil port as shown. When using the Unified Phantoms landmark the coil at the centre of the phantom that is away from the LPCA.

3.5 Procedure

- 3.5.1 To run the MCQA tool for the posterior coil, setup the posterior coil and phantom according to Table 3.1.
- 3.5.2 From the MR service desktop window, Image quality tab, select Multi-Coil QA Tool, as shown in Illustration 3.1.

Illustration 3.1: Select MCQA tool from the MR service desktop window.

3.5.3 On the test side, select “Click here to start the tool”

3.5.4 For further instructions on the MCQA tool, it can be found on the Discovery MR750 Service Methods CD or the web at: [Troubleshooting -> System -> Multi-Coil Quality Assurance Tool](#).

To run the MCQA tool for the anterior coil, setup the anterior coil and phantom according to Table 3.2. Repeat steps 3.5.2 – 3.5.4 on the anterior coil.

4 3.0T 32-Channel Torso Array Coil Troubleshooting Tips

4.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Timing Not Applicable	15 mins	Timing Not Applicable

4.2 Overview

The following tips can be used to troubleshoot common problems with the 3.0T 32-Channel Torso Array coil. These are some basic troubleshooting suggestions. The coil should not be replaced as a result of these checks alone. Check for other failures that would cause image quality issues and if additional help is needed, contact the online center at <http://3.87.118.28/optec3/common/mr750/frameset.htm>.

4.3 Preliminary Requirements

4.3.1 *Tools and Test Equipment*

Item	Qty	GE Part #
Digital Multimeter	1	46-194427P284
Legacy Phantom*	2	5341595 Alternate Part Numbers: 2381682-10 or 2406200 or 5325953-6
Unified Phantom*	2	5343347-2
Phantom Positioner	1	5339708-5

* Note: Required to use Legacy Phantom or Unified Phantom.

4.3.2 *Replacement Parts*

See Section 5: 3.0T 32-Channel Torso Array Coil Field Replaceable Units.

4.4 Receiving No Signal

4.4.1 *Receiving No Signal*

Problem:

You are unable to pre-scan or are scanning and receiving no signal or receiving partial signal from coil elements.

Possible Solution:

- Verify that the image on the P-connector is a green lock. This indicates that the connector is locked into place.
- Verify that the system has correctly detected the coil by checking in the “Currently Connected” in the GUI to select coils in Rx as shown in Illustration 4.1.
- Verify that the landmark is correct and that the cradle has not unlatched.
- Verify that the scan locations and any FOV offsets are correct.

- Perform a continuity check on the output cable (Section 4.6). The continuity check must be performed by a GE authorized Service Engineer.
- Verify that the coil is positioned with the cable exiting towards the bore.
- Verify that the cable is not looped or crossed.

There can also be a problem at the system interface port. Try scanning in the other ports. If you still cannot get a signal, try to scan (transmit and receive) with the body coil. For this test, be sure to remove the imaging coil from the magnet bore before you scan with the body coil. If you still receive no signal the problem probably lies with the MR system. If the scan completes successfully, there is probably a problem with the Torso coil. Contact GE for further assistance. If you are unable to scan with the substitute coil, there may be a system problem related to this particular coil type.

4.4.2 Image Quality

Problem:

Poor IQ, shaded images, or if MCQA fails

Possible Solution:

- Perform a continuity check on the output cable (Section 4.6). The continuity check must be performed by a GE authorized Service Engineer.
- If the check on the output cable shows that the cable is OK and the low SNR problem happens to one or more of the anterior coil channels, replace the anterior coil.
- If the check on the output cable shows that the cable is OK and the low SNR problem happens to one or more of the posterior coil channels, replace the posterior coil.
- Verify that there are no loops in the cables.
- Verify that there are no metal or ferromagnetic objects close to the coil, patient or magnet (i.e., safety pin, hair pin).
- Verify that the coil is properly positioned.
- Verify that your center frequency is within the frequency adjustment range for your system.
- Verify that the R1, R2 and TG values from the pre-scan are within normally expected ranges.
- Verify that no hair is trapped in the connectors at the system interface. This may appear as intermittent oscillations.

If you have not done so already, perform the coil Quality Assurance test (refer to section 3). If the values you obtain do not fall within normal operating parameters, investigate this further by performing a phantom scan with the body coil. For this test, be sure to remove the imaging coil from the magnet bore before you scan with the body coil. If you still have the same problems, there is probably an MR system problem. If the body coil scan is satisfactory, acquire a scan using another coil of the same type (receive-only, phased array). If the image quality is visibly improved, there may be a problem with the Torso coil. Contact GE for further assistance. If the image quality still suffers, there may be a system problem related to imaging with this type of coil.

4.4.3 Artifacts

Problem:

There is a black line, strip lines, or signal void on the image.

Possible Solution:

- Verify that there are no metal or ferromagnetic objects close to the coil, patient or magnet (i.e., safety pin, hair pin).
- Verify that no hair is trapped in the connectors at the system interface. This may appear as intermittent oscillations.
- Verify the cable connectors are fully engaged into the receptacle on the posterior and anterior sections of the unit. If the connector is not fully engaged, there may be a coil not present error or there may be intermittent oscillations.
- If you have oscillation of the preamp, you will see striped lines.

4.4.4 Output Cable Check

Visually inspect all coaxial connectors and pins on the system plug. If there are any broken, deformed or recessed pins or coaxial connectors, replace the cable assembly. For removal of the cable assembly, see Section 6.

4.5 RF Signal Channels Check

NOTE: The center pins of the System Connector, receive coaxial connectors, are extremely fragile. Do not attempt to make any measurements at these center pins.

The following will rule out a short to GND for each of the signal pins detailed in the following table 4.1. This check does not determine continuity for each of the MC signal connections.

Illustration 4.1: System Side P-connector as viewed from pin side of the connector.

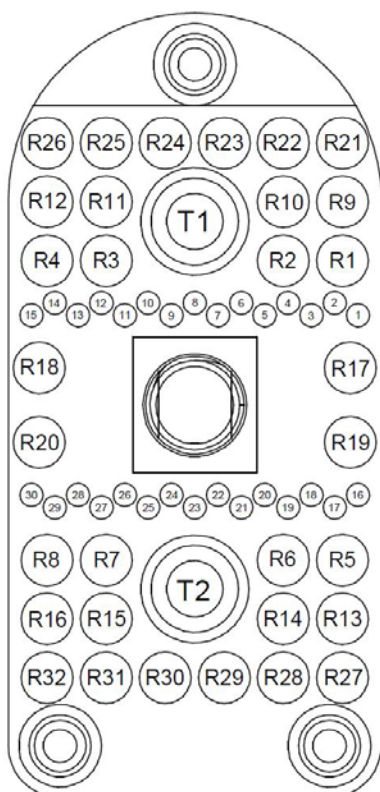
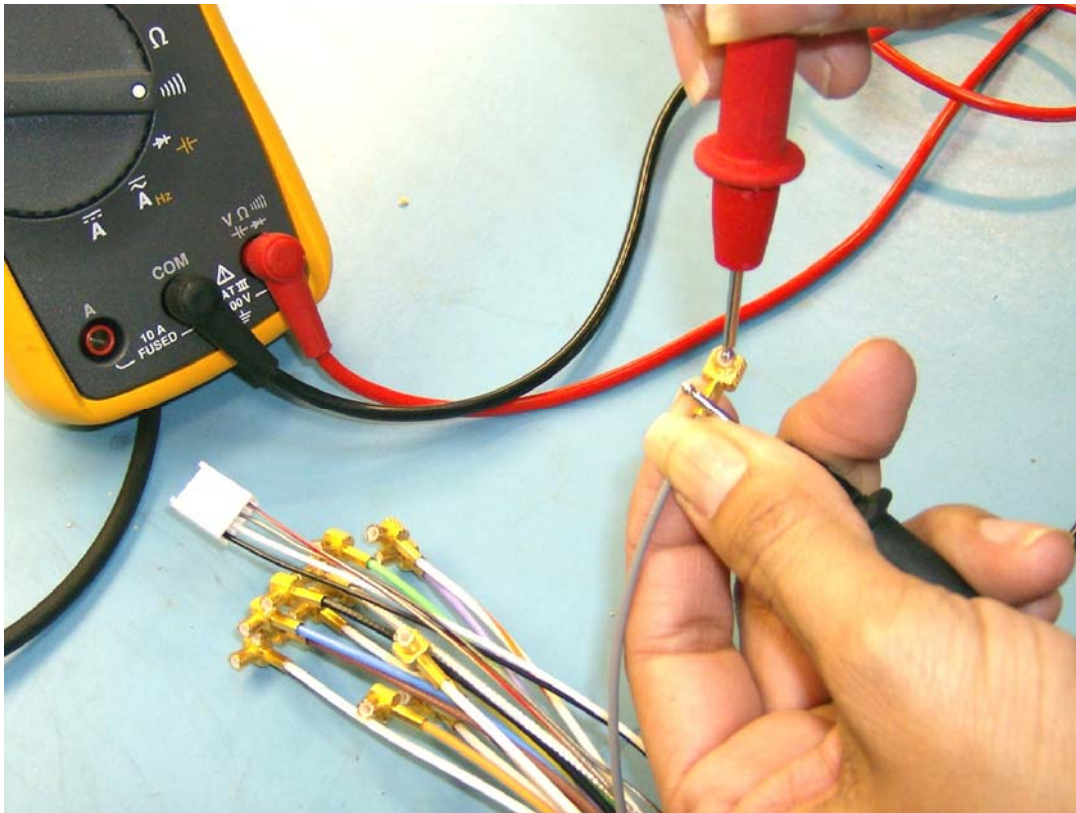


Illustration 4.2: Coil side connectors



With the coil cable disconnected from the coil (refer to 6.4.1.1 – 6.4.1.6 for instructions on how to remove the posterior coil and 6.4.2.1 – 6.4.2.6 for instructions on how to remove the anterior cable), use a multimeter (set to Continuity) to ensure an open circuit condition exists between the designated Coil side connector center pin and outer conductor shield for each signal channel shown in Table 4.1. Refer to Illustration 4.3 for an example of measuring the continuity for the RF contacts.

Illustration 4.3: Continuity measurement for RF contacts**Table 4.1:** System Cable RF wire connection locations

Coil Side Connector Color (RF GND Shield)	Coil Side Connector Color (RF Center Pin)
BROWN	BROWN
RED	RED
GREEN	GREEN
BLUE	BLUE
BLACK	BLACK
GRAY	GRAY
WHITE	WHITE
CLEAR	CLEAR
ORANGE	ORANGE
YELLOW	YELLOW
VIOLET	VIOLET
WHITE/BROWN	WHITE/BROWN
WHITE/RED	WHITE/RED
WHITE/ORANGE	WHITE/ORANGE
WHITE/YELLOW	WHITE/YELLOW
WHITE/BLACK	WHITE/BLACK

If one or more of the readings indicate a short, replace the cable. If all of the above readings indicate an open circuit condition, proceed to the DC Lines Continuity Check.

4.6 DC Lines Continuity Check

With the coil cable disconnected (refer to 6.4.1.1 – 6.4.1.6 for instructions on how to remove the posterior coil and 6.4.2.1 – 6.4.2.6 for instructions on how to remove the anterior cable), use a multimeter to determine continuity between the designated pins for each control channel. Refer to Table 4.2 for a list of connection locations. Illustration 4.1 shows the system side connector while Illustration 4.4 shows the DC connector and the locations of each pin. If any pair indicates an open, replace the cable. Refer to Illustration 4.5 for an example of measuring the continuity for the DC lines.

Table 4.2: System Cable DC wire connection locations

System Side Connector	Coil Side Connector
5	J1-1
6	J1-2
18	J1-3
20	J1-4
SHORT 1 to 2	
SHORT 3 to 4	

Illustration 4.4: DC connector J1

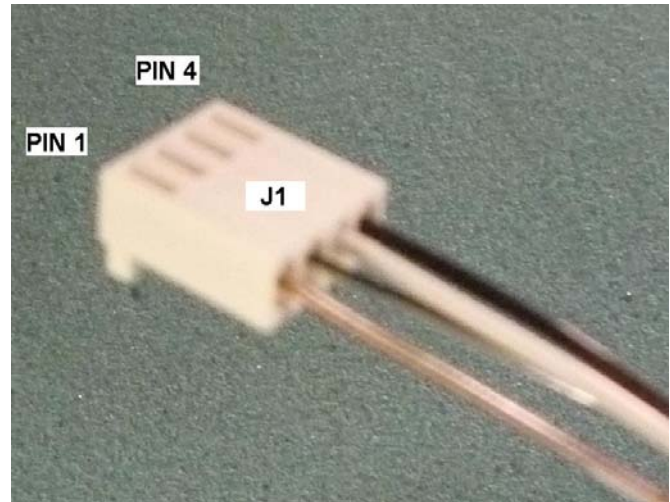
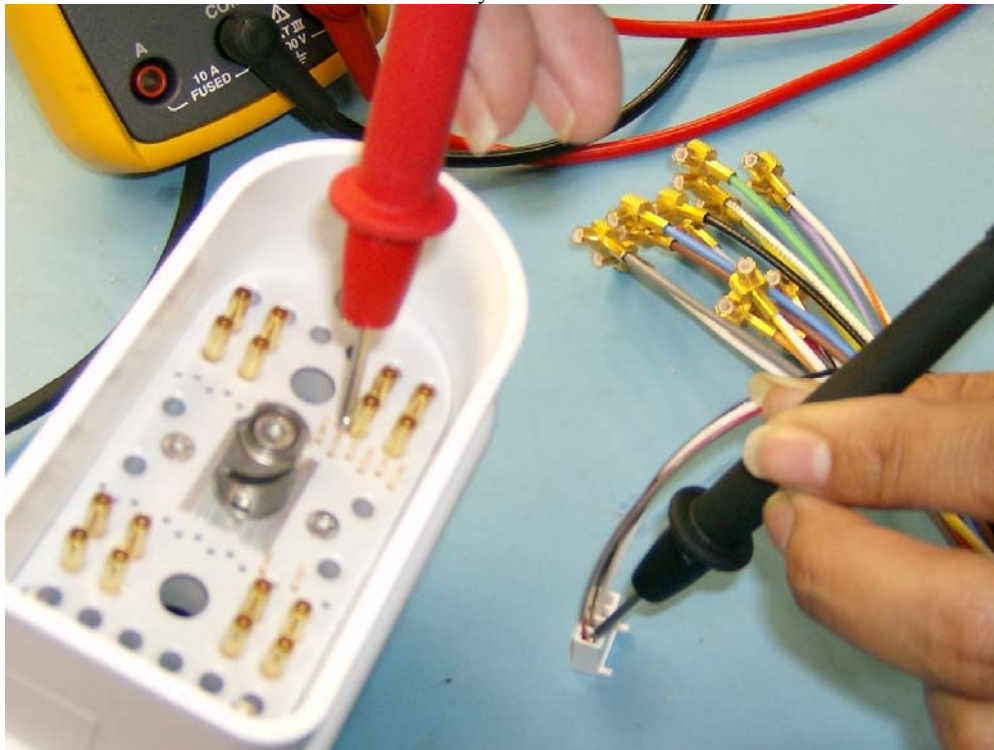


Illustration 4.5: Continuity measurement for DC contacts



5 3.0T 32-Channel Torso Array Coil Field Replaceable Units

Table 5.1: FRUs List

GE Part No.	Description
5339708-2	3T 32Ch Torso Coil - Anterior with Cable
5339708-3	3T 32Ch Torso Coil - Posterior with Cable (shipping also includes straps – NeoCoil P/N NC004034)
5339708-4	3T 32Ch Torso Coil - System Cable
5339708-5	3T 32Ch Torso Coil - Posterior Phantom Positioner
5339708-6	3T 32Ch Torso Coil - Spare Kit (see table below)
5341595 Alternate Part Numbers: 2381682-10 or 5325953-6	G3 Large Brick Phantom (Legacy)
5343347-2	Unified TL Brick Phantom

Table 5.2: Spare Kit Contents (GEHC Part # 5339708-6)

NeoCoil Part No.	Quantity	Description
NC00M067-4	4	Skid-Pads, Viton
NC00M023	1	Clamp, System Cable
NC00M001-43	10	4mm X 12mm Brass Pan-Head Screw
NC00M001-44	10	4mm X 16mm Brass Pan-Head Screw
NC00M079	10	Screw Cover Caps
NC00M080	10	Screw Cover Bases
NC00M020-4	20	Washers
NC00M084	2	Clip, Cable
NC004091	1	Instruction Sheet

Table 5.3: Other Replaceable Accessories List

ACCESSORIES MUST BE ORDERED BY CUSTOMER. NOT FRUS. NOT COVERED UNDER SERVICE CONTRACT.

NeoCoil Part No.	Description
NC004034	Straps, Posterior
NC004012	Pad Kit (all pads listed below purchased as a kit, all pads can be purchased separately)
NC004006	Chin Pad
NC004007	Anterior Patient Comfort Pad
NC004008	Cable Cover Pad
NC004033	Posterior Patient Comfort Pad

6 3.0T 32-Channel Torso Array Coil Cable Replacement

6.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Timing Not Applicable	10 mins	10 min

6.2 Overview

Follow this process to change the system cable for coils that have a cable available as a replaceable part.

6.3 Preliminary Requirements

6.3.1 *Tools and Test Equipment*

This procedure requires a standard toolkit.

6.3.2 *Replacement Parts*

Table 6.1: 3.0T 32-Channel Torso Array replacement parts

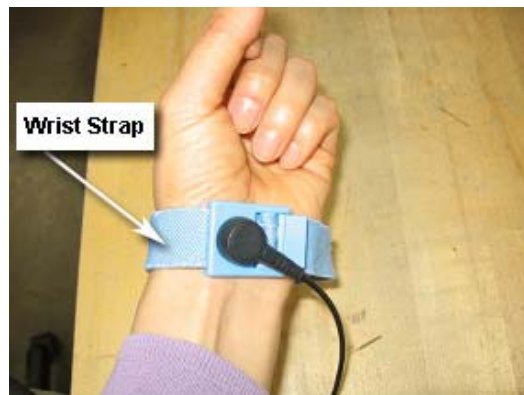
Item	Qty	Part#
System Cable	1	5339708-4
Spare Parts Kit (recommended during FRU replacement in case parts are lost)	1	5339708-6

6.4 Procedure



Observe anti-static precautions: Use wrist strap and ESD workstation (ESD mat connected to ground and place to connect ESD wrist strap).

Illustration 6.1: ESD Wrist Strap On



6.4.1 Replacing the Posterior System Cable

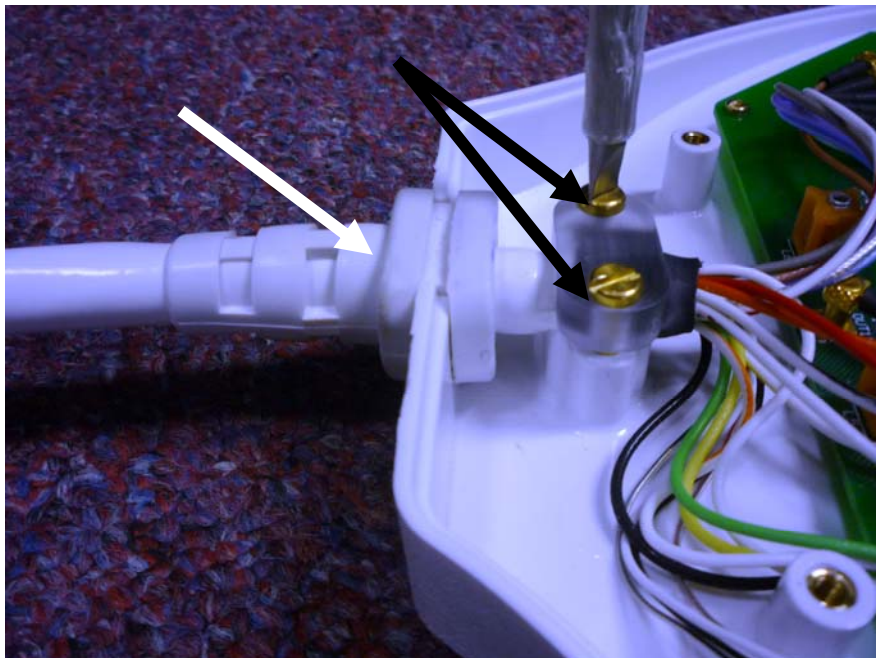
- 6.4.1.1 Put an ESD wrist strap on and ground.
- 6.4.1.2 If applicable, remove the straps from the posterior assembly.
- 6.4.1.3 Unscrew the bottom cover and retain the original sixteen (16) screws used. See Illustration 6.2.

Illustration 6.2: Removing the Posterior cover



- 6.4.1.4 Remove the posterior cover.
- 6.4.1.5 Disconnect all MCX connectors and the DC connector J1. It may be necessary to use pliers to remove the MCX connectors.
- 6.4.1.6 Unscrew the cable clamp from the coil as shown in Illustration 6.3. Retain the original screws used. Lift cable from housing to remove.

Illustration 6.3: Removing the Cable Clamp



- 6.4.1.7 Replace all MCX connectors, using Table 6.2 for the locations of each connector

Table 6.2: System Cable MCX connector location guide

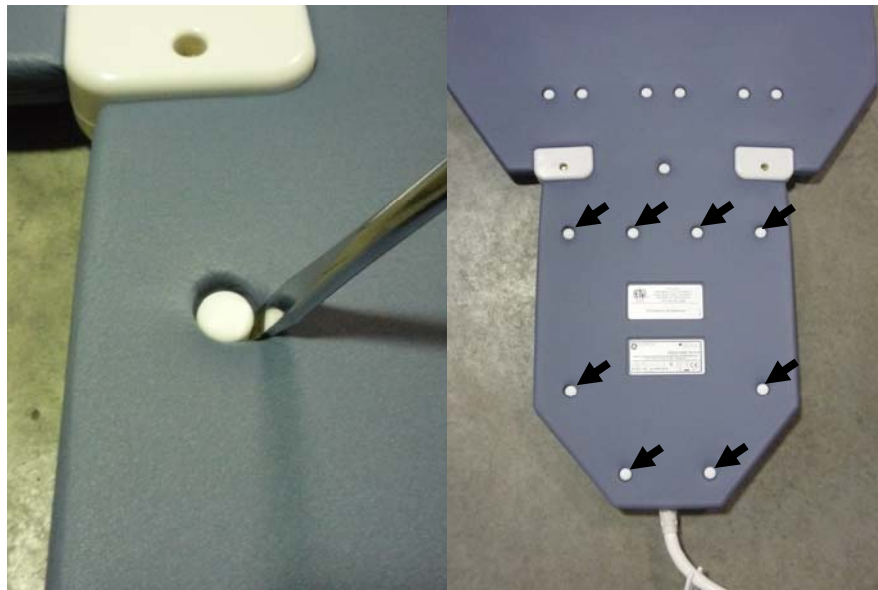
PCB Connector Reference Designator	MCX Connector Cable Color
OUT1	BROWN
OUT2	RED
OUT3	GREEN
OUT4	BLUE
OUT5	BLACK
OUT6	GRAY
OUT7	WHITE
OUT8	CLEAR
OUT9	ORANGE
OUT10	YELLOW
OUT11	VIOLET
OUT12	WHITE/BROWN
OUT13	WHITE/RED
OUT14	WHITE/ORANGE
OUT15	WHITE/YELLOW
OUT16	WHITE/BLACK

- 6.4.1.8 Slide the strain relief (white arrow) into the Posterior Housing ensuring that the flat edge is flush with the housing. Refer to Illustration 6.3.
- 6.4.1.9 Secure the cable using the cable clamp and two (2) screws. It is highly recommended to torque the screws to 8in-lbs. Do not over tighten.
- 6.4.1.10 Replace the Posterior bottom cover. Use original screws to fasten cover. It is highly recommended to torque the screws to 8in-lbs. Do not over tighten.

6.4.2 Replacing the Anterior System Cable

- 6.4.2.1 Put an ESD wrist strap on and ground.
- 6.4.2.2 Remove the eight (8) screw covers using a screw driver to pry them up. See Illustration 6.4.

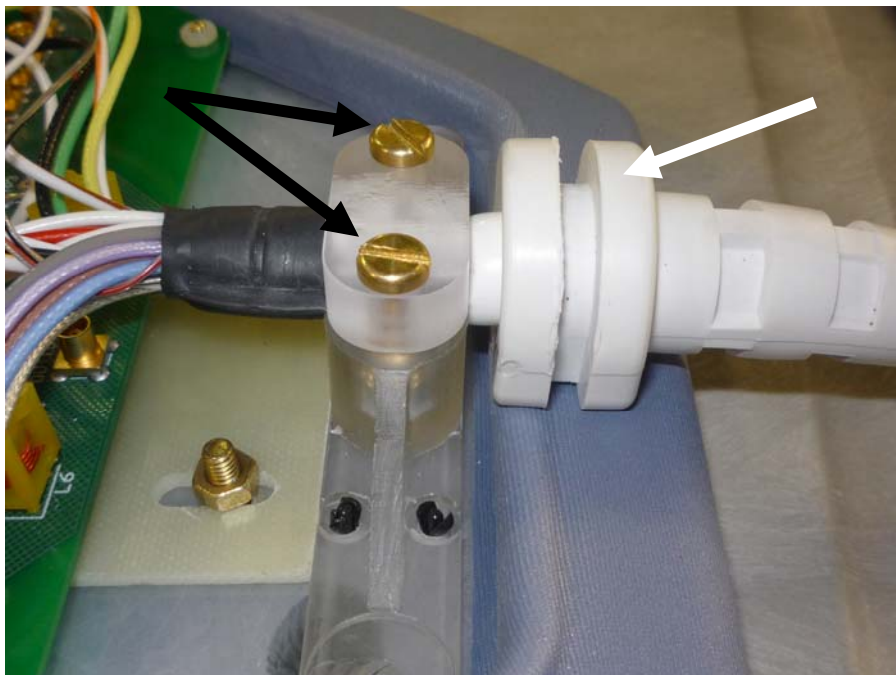
Illustration 6.4: Anterior screw covers



- 6.4.2.3 Unscrew the preamp cover and retain the original eight (8) screws, cap bases and washers (two per). Flip over the coil and lift off the preamp cover. See Illustration 6.5.

Illustration 6.5: Anterior preamp cover

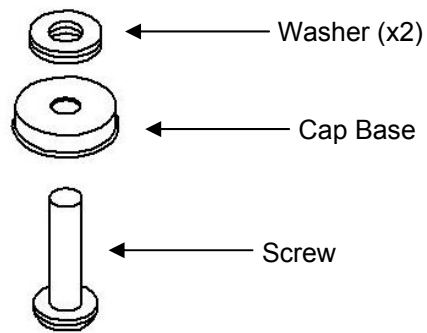
- 6.4.2.4 Disconnect all MCX connectors attached to the system cable and the DC connector J1. It may be necessary to use pliers to remove the MCX connectors.
- 6.4.2.5 Unscrew the cable clamp (2 screws) from the coil as shown in Illustration 6.6. Retain the original screws used and clamp.
- 6.4.2.6 Remove the original system cable.

Illustration 6.6: Cable Clamp and Strain Relief location

- 6.4.2.7 Replace the cable and re-connect all MCX connectors, using Table 6.2 for the locations of each connector.

- 6.4.2.8 Position the strain relief (white arrow) on the Anterior foam, flush with the edge of the foam lip as shown in Illustration 6.6. Ensure that the flat edge faces downwards.
- 6.4.2.9 Secure the cable using the cable clamp and two (2) screws. It is highly recommended to torque the screws to 8in-lbs. Do not over tighten.
- 6.4.2.10 Replace the Anterior preamp cover using the original screws, cap bases and washers to fasten the cover (see Illustration 6.7 for the correct order of parts). It is highly recommended to torque the screws to 8in-lbs. Do not over tighten.

Illustration 6.7: Screw, cap base and washers order.



- 6.4.2.11 Snap the screw covers back into place using your thumb.

6.5 Finalization

Perform a final MCQA test with this coil to confirm that the repaired coil operates properly. Refer to Section 3.

Revision History

Rev	Date	Author	Primary Reason For Change
1	5/18/2009	Erik von Asten	Initial Revision
2	6/24/2009	Erik von Asten	Updated dimensions of the coil in section 1.5. Added alternate phantom part numbers to sections 2.2.1, 3.2.1, 4.3.1, and Table 5.1.
3	7/29/2009	Erik von Asten	Clarifications to Illustration 1.2 not showing straps and pads. Chapter 2 was updated with a new installation procedure. Added Illustration 3.2, 4.3 and 4.5. Updated wording in sections 4.4.2, 4.5, 4.6, 6.4.1.5, 6.4.1.6, 6.4.2.3, 6.4.2.4, 6.2.4.5, 6.4.2.8. Added sections 3.5.4, 3.5.13, 6.4.1.2, 6.4.1.4, 6.4.2.6. Updated tables 5.1 and 6.3.2. Deleted blank page after section 5.
4	3/10/2010	Erik von Asten	Added Unified Phantom set to sections 2.2.1, 3.2.1, 4.3.1 and Table 5.1. Updated section 2.3.1. Added pictures of the Unified phantoms to Table 3.1 and 3.2. Updated MCQA section with a reference to the new procedure online. Added Illustration 6.7. Added reference to cap base and washers in section 6.4.2.3 and 6.4.2.10. Added washers to Spare Parts Kit.